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Multiple stressors affecting microbial decomposer and litter decomposition in restored urban streams: Assessing effects of salinization, increased temperature, and reduced flow velocity in a field mesocosm experiment

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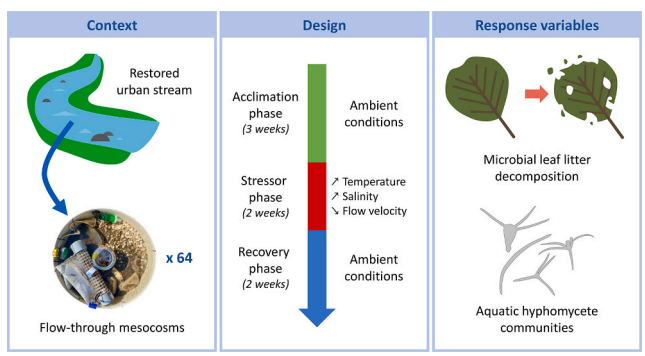
HIGHLIGHTS

- Response of restored urban stream ecosystems to multiple anthropogenic stressors
- Flow-through mesocosms to test for stressor impacts on litter decomposition
- Small impact of warming and reduced flow velocity, very limited impact of salinity
- Rare and weak interactive effects of flow conditions, salinity, and warming
- High resistance of urban fungal decomposers in restored streams to modest stress

GRAPHICAL ABSTRACT



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ABSTRACT

A multitude of anthropogenic stressors impact biological communities and ecosystem processes in urban streams. Prominent among them are salinization, increased temperature, and altered flow regimes, all of which can affect microbial decomposer communities and litter decomposition, a fundamental ecosystem process in streams. Impairments caused by these stressors individually or in combination and recovery of communities and ecosystem processes after release from these stressors are not well understood. To improve our understanding of multiple stressors impacts we performed an outdoor stream mesocosm experiment with 64 experimental units to assess the response of microbial litter decomposers and decomposition. The three stressors we applied in a full-factorial design were increased salinity (NaCl addition, 0.53 mS cm^{-1} above ambient), elevated temperature (3.5 °C above ambient), and reduced flow velocity (3.5 vs 14.2 cm s^{-1}). After two weeks of stressor exposure (first sampling) and two subsequent weeks of recovery (second sampling), we determined leaf-associated microbial respiration, fungal biomass, and the sporulation activity and community composition of aquatic

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hyphomycetes in addition to decomposition rates of black alder (*Alnus glutinosa*) leaves confined in fine-mesh litter bags. Microbial colonization of the litter was accompanied by significant mass loss in all mesocosms. However, there was little indication that mass loss, microbial respiration, fungal biomass, sporulation rate or community composition of aquatic hyphomycetes was strongly affected by either single stressors or their interactions. Two exceptions were temperature effects on sporulation and decomposition rate. Similarly, no notable differences among mesocosms were observed after the recovery phase. These results suggest that microbial decomposers and leaf litter decomposition are either barely impaired by exposure to the tested stressors at the levels applied in our experiment, or that communities in restored urban streams are well adapted to cope with these stressor levels.

1. Introduction

Urban streams are facing strong anthropogenic pressures that affect aquatic communities and ecosystem functioning (Booth et al., 2016; Grimm et al., 2008). Among the suite of stressors commonly encountered in streams of urban catchments are warming resulting from local heat inputs (Zhao et al., 2014) and flow alterations (Walsh et al., 2016), which are superimposed by climate change (Hale et al., 2016; Zhao et al., 2021). Urban streams are also exposed to salinization caused by urban weathering, sewage effluents, and street salting (Kaushal et al., 2021). Furthermore, in industrial areas, mining activities often result in elevated salinity levels caused by tailing waters (Cañedo-Argüelles, 2020; Cunillera-Montcusí et al., 2022). Although different ions can contribute to salinization (Sauer et al., 2016; Cañedo-Argüelles, 2020), NaCl inputs are generally the dominant cause of increased salinity in urban streams (Kaushal et al., 2021; Utz et al., 2022). Given the potential for numerous interactions, impacts of multiple stressors on stream communities and ecosystem processes can be complex (Birk et al., 2020; Orr et al., 2020; Spears et al., 2021). This includes consequences for leaf-litter decomposition as a fundamental process determining stream ecosystem functioning (Follstad Shah et al., 2017; Marks, 2019), and for the microbial decomposers involved in the process, notably a polyphyletic group of fungi known as aquatic hyphomycetes (Gessner et al., 2007; Krauss et al., 2011; Kuehn, 2016).

Although elevated salinity has deleterious effects on freshwater communities (Cañedo-Argüelles, 2020), various studies have indicated that aquatic hyphomycetes are remarkably tolerant to salt pollution (Canhoto et al., 2021). For example, salinity levels well above the freshwater threshold of 1 ‰ were found to be necessary to elicit a response of several species grown on agar plates (Canhoto et al., 2017). Nevertheless, fungal communities can still be affected by elevated salinity in natural environments (Sauer et al., 2016; Oliveira et al., 2021; Schäfer et al., 2012). Moreover, salinization often acts in combination with other stressors (Stirpe et al., 2017), and these interactions may be a major reason why inconsistent responses of aquatic hyphomycetes to elevated NaCl levels have been observed (Gonçalves et al., 2019).

Hydromorphological modifications of streams (Berger et al., 2019) can be one such stressor that leads to complex interactions with increased salinity. For example, flow reductions modify habitat, including for microbial communities, by altering hydraulic conditions and favoring sediment deposition (Bastias et al., 2020), with negative consequences for fungal biomass and sporulation in experiments conducted in both experimental stream channels (Ferreira and Graça, 2006; Sanders and Webster, 1980; Schlief and Mutz, 2009) and the field (Colas et al., 2016). Potential mechanisms behind such impacts on aquatic hyphomycetes are a lack of stimulation of sporulation by water turbulence (Webster and Towfik, 1972), reduced attachment success due to altered leaf surface properties (Dang et al., 2007), or changes in oxygen or nutrient supply (Dewson et al., 2007; Gomes et al., 2018).

Increased temperature, in contrast, accelerates aquatic hyphomycete growth (Suberkropp, 1984; Dang et al., 2009), sporulation (Chauvet and Suberkropp, 1998) and biomass accrual (Ferreira and Chauvet, 2011a) and can also increase microbial litter decomposition (Ferreira et al., 2014; Martínez et al., 2020). However, temperature increases can also reduce fungal diversity and thereby modify community composition (Bärlocher et al., 2008; Fernandes et al., 2012), with potential effects on overall fungal community activity and litter decomposition. Given the curvilinear response of aquatic hyphomycetes to temperature (Dang et al., 2009) and seasonal succession patterns (Gessner et al., 1993), impacts of warming are also likely to vary with season (Ferreira and Canhoto, 2015).

While effects of multiple stressors on stream ecosystems have received considerable attention (e.g. Jackson et al., 2016; Birk et al., 2020), including on aquatic hyphomycetes and leaf litter decomposition (e.g. Bruder et al., 2016; Juvigny-Khenafou et al., 2020; Gonçalves et al., 2019; Madge Pimentel et al., 2024), little is known about stressor interactions and especially on responses when streams are being released from stressors. Gaining information on these responses is important both from a fundamental perspective (Vos et al., 2023) and in the light of massive investments in stream and catchment restoration (Gerner et al., 2018; Kroll et al., 2019) that require evaluating and monitoring success of the measures taken (dos Reis Oliveira et al., 2020; Palmer et al., 2005). While responses have been studied for macroinvertebrate (Staentzel et al., 2019), macrophytes (Lorenz et al., 2012), and fish (Roni, 2019), assessments of microbial communities and ecosystem processes remain limited (Hilderbrand et al., 2023), particularly in urban streams.

In the present study, we performed an outdoor flow-through mesocosm experiment to assess the effects of, and recovery from, multiple stressors and their interactions on microbial leaf litter decomposition and decomposers. Specifically, we tested whether leaf litter decomposition and litter-associated fungi and other microbes in urban streams respond to (1) elevated salinity, despite considerable tolerance of aquatic hyphomycetes to salinity and high background levels in urban streams; (2) reduced flow velocity because of limited fungal litter colonization; (3) temperature increases due to stimulated fungal activity and possibly a shift in community composition; (4) combinations of these stressors resulting from interactions; and (5) release from these stressors as a result of recovery of microbial litter decomposers.

2. Material and methods

2.1. Study site

The experiment was conducted in stream-side flow-through mesocosms at the Boye River (latitude: 51.5533° N, longitude: 6.9485° E), a tributary of the Emscher River located in the Rhine-Ruhr metropolitan region of Germany. Decades of wastewater discharge, coal mining and heavy industrial activities in the area had led to severe hydromorphological stream modifications and pollution, including the use of stream channels as open sewers (Gerner et al., 2018). Historical data also reveal strong salt pollution of the Boye River in the past (up to 15,500 mg/L chloride and a conductivity as high as 20 mS cm^{-1}; Madge Pimentel et al., 2024). A multi-billion-euro restoration project initiated in 1992 aimed at improving water quality and reshaping river morphology (Winking et al., 2014; EGLV, 2022). As a result, the Boye stream network has been free of sewage since 2017, and improvements of channel morphology across the entire network were also largely

completed over the last 20 years (Gillmann et al., 2023). Chloride concentrations have also decreased, although they can still be high (26–286 mg/L since 2010), along with conductivity (0.4–1.93 mS cm^{-1}; Madge Pimentel et al., 2024). During the experiment, the Boye River had a mean chloride concentration of 36.9 ± 3.6 mg/L at the experimental site (Table S1).

2.2. Experimental set-up

The experiment was conducted in an *ExStream* experimental system (Piggott et al., 2015) consisting of 64 circular outdoor flow-through mesocosms. Water was constantly supplied from the adjacent Boye River to ensure realistic physio-chemical conditions and colonization of the mesocosms by drifting microbes and macroinvertebrates. Two centrifugal pumps (Pedrollo NGAm 1A-Pro, maximum capacity: 315 L/min) supplied stream water to the mesocosms. A 5-mm mesh at the inlet retained suspended coarse material. The water was pumped to the top of a two-level scaffold (upper level 3 m, lower level 1 m above ground), from where it flowed to two serially connected settling tanks (203 L rain barrels) to facilitate sedimentation of suspended material. Water from the settling tanks fed four header tanks (203 L rain barrels), each of which supplied a block of 16 mesocosms. The circular mesocosms (outer diameter: 25 cm, central outflow: 6 cm, volume 3.5 L; Fig. 1A) on the lower level of the scaffold received water through a 90° L-piece. This produced a clockwise flow in the mesocosms before the water left through the central outlet (Fig. 1A). The outflow and any surplus water from the settling and header tanks were directed back to the Boye River downstream from the intake site.

The experiment involved an acclimation phase (March 4 – March 24, 2022), a stressor phase (March 24 – April 7, 2022), and a recovery phase (April 7 – April 21, 2022). During the acclimation phase, all mesocosms received untreated stream water and were colonized by drifting organisms (Fig. 1B). On acclimation day one, the mesocosms were supplied with substratum reflecting the composition of the Boye stream bed. Sand was the dominant substratum in the stream, with larger grain sizes occurring only occasionally (http://www.elwasweb.nrw.de, accessed on 6 November 2023). Thus, 1 L sediment per mesocosm was collected near the experimental site (latitude: 51.5545°, longitude: 6.9464°), sieved and the size fraction <1 mm thoroughly mixed with a slurry of 100 mL fine particulate organic matter (FPOM) collected from a nearby tributary (latitude: 51.5627°, longitude: 6.9154°) to boost microbial inoculation of the mesocosms. Fine gravel (6–8 mm grain size) purchased in a hardware store was used to stabilize the fine sediment in the first quarter of the mesocosms immediately below the inlet. Four pieces of standardized coarse gravel (3 × 40–80 mm, 1 × 32–56 mm) also from a hardware store were placed on top of the sediment at the two ends of the

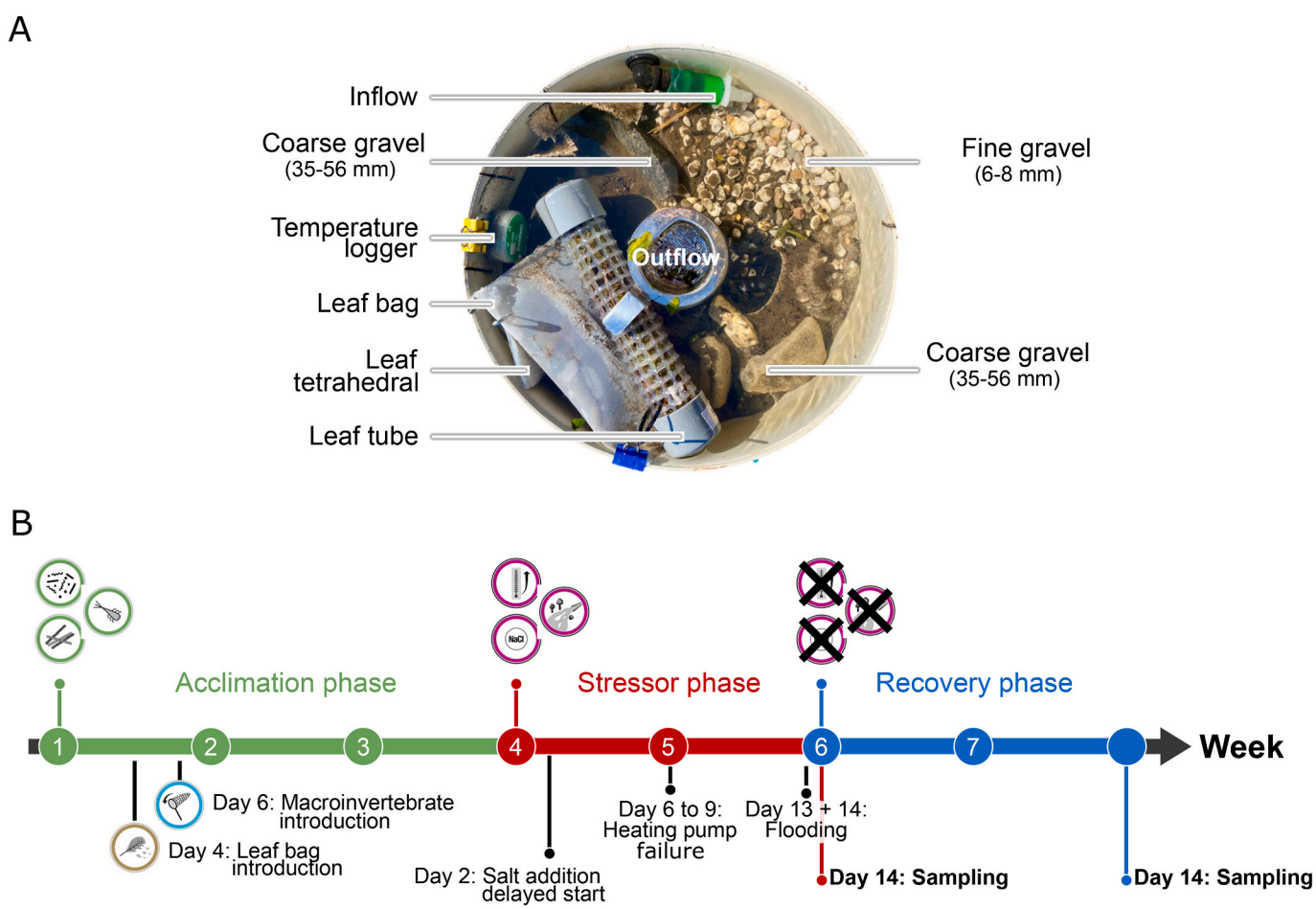

Fig. 1. Overview of the ExStream experiment. A: Design of the circular mesocosms. The mesocosms were constantly supplied with water through the inflow, and the water left through the outflow. The substrate in the mesocosm corresponded to the dominating substrate in the Boye River (fine sediments, gravel, stones). Each mesocosm was supplied with leaf litter, enclosed in a coarse-mesh tube ("leaf tube") and in two fine-mesh bags ("leaf bag" and "leaf tetrahedral"). In this study, the focus is only on the "leaf tetrahedral" bag. During the stressor phase, for the corresponding treatments, (i) warm water was introduced through the inflow, (ii) reduced flow velocity was achieved by removing the white removable inflow jet and (iii) salt was added using an external dripper line. B: Timeline of the experiment, consisting of three phases: (i) acclimation (20 days – 64 mesocosms) in which the mesocosms were supplied with untreated stream water and were colonized by organisms; (ii) stressor phase (14 days – 64 mesocosms) where the three stressors were applied in a full-factorial design with 8 replicates per treatment combination; (iii) recovery phase (14 days – 32 mesocosms) where the stressor treatments were discontinued.

fine gravel section (Fig. 1A). Discharge in the mesocosms was adjusted two to three times a day to approx. 2.1 L min^{-1} by manually closing or opening shut-off valves.

Leaf litter from black alder (*Alnus glutinosa* (L.) Gaertn.) collected in autumn 2021 and stored air-dried, was introduced into the mesocosms on acclimation day four to determine rates of microbial litter decomposition and analyze the litter-associated microbial communities. Specifically, 12 discs (10 mm in diameter) were cut from remoistened leaves with a cork borer, lined up on a safety pin, individually separated by small beads, and placed in tetrahedral fine-mesh bags (0.5 mm mesh size, 5 cm edge length; Fig. 1A), which were subsequently sealed to prevent feeding on the leaf material by macroinvertebrates.

2.3. Stressor exposure

During the 14-day stressor phase, three stressors were applied in a full-factorial design with four-fold replication. To induce salinity stress in randomly selected mesocosms within each block, a saturated NaCl stock solution (>350 mg L^{-1}) was prepared in a tank (REKUBIK 1000 L IBC container) by dissolving refined salt tablets (>99.9 % NaCl, Claramat®) in stream water that was constantly circulated by a submersible pump (9000Dirt, Gardena). Dosage pumps (GHL Doser 2.1) delivered the concentrated salt solution to a water supply line, which also received filtered stream water (100 μm mesh size) and then released the diluted salt solution through pressure-compensated drippers to the salt-receiving mesocosms at a rate of 4 L h^{-1}. Erroneous assignment of one salinization treatment resulted in five instead of four replicate mesocosms for the combination of reduced flow velocity with warming, and for salinization combined with warming. Conversely, the treatment involving warming only and all three stressors combined had only three replicates instead of four. This resulted in an unbalanced design for the stressor phase.

Electrical conductivity (EC) was measured once daily in all mesocosms to monitor salinity levels (Cond 315i, WTW, Weilheim, Germany). Whenever significant flow reductions in mesocosms were noticed during flow calibration, conductivity was measured before re-adjusting the discharge to identify any short-term salinity peaks (Table S2). The salt addition was stopped for 1.5 days on day 13 to prevent system clogging after heavy rainfall. This resulted in 11 instead of 14 days of elevated salinity. The average increase in conductivity above ambient was 0.53 mS cm^{-1} (mean ± SD: $EC_{ambient} = 0.84 \pm 0.01$ mS cm^{-1}, $EC_{saline} = 1.34 \pm 0.15$ mS cm^{-1}), corresponding to 154 mg L^{-1} of added chloride. Short-term salinity peaks resulting from flow rate reductions are not considered in this calculation. According to a projection for rivers worldwide, the median electrical conductivity is estimated to increase by 0.2 mS.cm^{-1} by the end of the century.

Warming was applied block-wise to a total of 32 mesocosms. To this end, temperature in two of the four header tanks was increased by 4.5 °C (± 0.2 °C) above ambient to achieve a 3.5 °C temperature increase above ambient by the time the water reached the mesocosms. Centrifugal pumps (Onga 400 Series, Melbourne, Vic, Australia) fitted with a 1-mm mesh at the inlets supplied a warm and an ambient water line, respectively. A self-cleaning filter system (120 μm mesh size) prevented larger particles from entering (Madge Pimentel et al., 2023). A petroleum-powered mobile heating system (350 kW; Mobile Wärme 24 GmbH, Wuppertal, Germany) was used to raise the temperature in the warm water line to 50 °C. Warm and ambient water was supplied to a mixing module between the second settling tank and the header tank by using an automated modular heating system (Madge Pimentel et al., 2023) enabling rapid mixing of cold and warm stream water to reach the target water temperature while minimizing exposure of drifting organisms to high temperatures. Heating was interrupted for 3.5 days after day 6 because a pump failed, and this happened once again for 1.5 days on day 13. Both interruptions were due to high sediment load in the river, and resulted in 9 instead of 14 days of experimental warming. Temperature in two randomly selected mesocosms per block was recorded every 5 min using HOBO Pendant MX temperature/light data loggers (Onset, Bourne, MA, USA). During the stressor application phase, the average increase in water temperature between heated and ambient mesocosms was 3.5 °C (mean ± SD: $T_{ambient} = 8.7 \pm 0.1$ °C, $T_{heated} = 12.2 \pm 0.1$ °C; $N = 4$), which compares with observed increases in water temperature in rivers of up to 1 °C per decade (Parmesan et al., 2022), and is similar to the level of warming applied in other experiments on litter decomposition and decomposers in streams (e.g. Bärlocher et al., 2008; Ferreira and Canhoto, 2015).

Like salinity stress, flow velocity levels were reduced in randomly selected mesocosms within each block. This was achieved by removing the nozzle of the inlet pipe while keeping the water volume passing through the mesocosms at constant level. The reduced flow velocity measured at mid-depth in the first quarter of the mesocosms was 3.5 ± 3.3 cm s^{-1} (N = 4), compared to the reference flow velocity of 14.3 ± 7.6 cm s^{-1} (N = 4). This flow velocity reduction also decreased flow variability in the mesocosms by eliminating fast-flowing sections, as has been observed in urban streams (Konrad, 2005; Table S3).

2.4. Sampling

Half of the mesocosms exposed to each stressor combination were sampled 14 days after the start of the stressor phase. The other half was released from stressors (nozzle reattached to inlet pipes, warming and salt addition discontinued) and sampled after an additional 14 days to allow for system recovery (Fig. 1B). Litter bags were collected from 32 mesocosms at the end of the stressor phase, and the remaining 32 mesocosms at the end of the recovery phase. The samples were placed in separate boxes with stream water mimicking the prior salt exposure conditions and immediately processed upon return to the laboratory. The leaf discs were carefully removed from the litter bags, cleaned with unfiltered stream water, and briefly stored submerged at field temperature in a climate chamber before measuring microbial respiration rates.

2.5. Microbial respiration

Two leaf discs per litter bag were used for duplicate measurements of oxygen consumption in a total of four glass microplates, each accommodating 24 wells (1.7 mL) equipped with individually calibrated O_2 sensor spots (Loligo Systems, Viborg, Denmark). All wells of a microplate were filled with filtered stream water (47-mm-diameter Whatman GF/C filter), the sensor spot was covered with a fine mesh for protection, and a metal bead was added together with a leaf disc to ensure efficient mixing within the wells. Care was taken to avoid trapping air bubbles when sealing the wells. Background O_2 consumption was measured in eight wells filled only with filtered stream water. The microplates were placed in a climate chamber at 12 °C on O_2 fluorescence readers (SDR SensorDish Reader, PreSens, Regensburg, Germany) submerged in a water bath at 12 °C, corresponding to the intermediate temperature between the unheated control and heated mesocosms the day before retrieving the litter bags. The microplates were also placed on an orbital shaker to ensure tight temperature control and thorough mixing of the water in the wells. Maintaining a constant incubation temperature for respiration and sporulation (see below) measurements ensured standard conditions for comparisons of the different treatment combinations in the mesocosms.

2.6. Aquatic hyphomycete sporulation and community composition

To determine sporulation rates and the community composition of sporulating aquatic hyphomycetes, all cleaned leaf discs, including those used for respiration measurements, were placed in Petri dishes filled with 15 mL of filtered stream water and incubated on an orbital shaker (50 rpm) at 12 ± 1 °C and a photoperiod of 4 h light, 12 h dark and 8 h light for a total of 24 h. The resulting spore suspension was decanted, the leaf discs and Petri dish rinsed with 4 mL of filtered stream water, and

the combined suspension fixed with 1 mL of 37 % formalin for later microscopic analysis of the spores. The fixed spore suspensions were gently mixed to ensure a homogeneous spore distribution before filtering a 5-mL aliquot on a 5-μm-pore size cellulose nitrate membrane filter (Whatman) and staining the spores with 0.1 % trypan blue in 60 % lactic acid (Gessner et al., 2003). The spores were counted and identified in 50 randomly chosen microscopic fields per filter (200× magnification), using primarily the key by Gulis et al. (2020) for identification. Sporulation rates were expressed as the number of spores released per mg of dry mass per day. Richness of the sporulating aquatic hyphomycetes was expressed as the number of species identified per mesocosm.

2.7. Litter mass loss

Following the respiration measurements and the incubations to induce fungal sporulation, the leaf discs were stored at −20 °C, and later freeze-dried and weighed to determine the percent dry mass lost by subtracting the final dry mass remaining from the estimated mean initial dry mass of leaf discs not submerged in the mesocosms (62.6 mg ± 3.6, $N = 6$). Decomposition rate coefficients, k, were calculated by assuming exponential decay according to the model $M_t = M_0\, e^{-kt}$, where M_0 is the initial litter mass and M_t is the litter mass remaining at time t (i.e. 38 days after the stressor phase, and 53 days after the recovery phase). Decomposition rates were also quantified after t degree days to normalize for any variation in temperature among mesocosms.

2.8. Fungal biomass

Ergosterol was extracted and quantified to estimate fungal biomass (Gessner, 2020). Briefly, five freeze-dried discs were used to extract lipids and saponify esters for 30 min at 80 °C in 10 mL of alkaline methanol (8 g KOH/L methanol). The resulting extract was cleaned by solid-phase extraction (SPE; Sep-Pak Vac-RC-500 mg, Waters, Milford, MA, USA). Ergosterol was purified and quantified by high-performance liquid chromatography (HPLC) using an Infinity II 1260 chromatograph (Agilent Technologies, Santa Clara, CA, USA), a LiChrospher 100-5RP 18 column (Merck, Darmstadt, Germany) maintained at 33 °C, with pure methanol as the mobile phase at a flow rate of 1.4 mL min^{-1} and ergosterol detection at 282 nm. Fungal biomass was expressed as μg ergosterol per g litter dry mass.

2.9. Statistical analyses

All analyses were performed in *R* version 4.1.2 (R Core Team, 2021). Five response variables were chosen to assess effects on microbial leaf litter decomposers and decomposition: fungal sporulation rate, fungal species richness, fungal biomass, microbial respiration, and litter decomposition rate, using the linear model:

$$Y = \alpha + \beta_1 S + \beta_2 V + \beta_3 T + \beta_4 SV + \beta_5 ST + \beta_6 VT + \beta_7 SVT + \varepsilon, \quad (1)$$

where Y is either of the response variables, T, V, and S are the vectors corresponding to temperature, flow velocity, and salinity respectively; α and β are the estimated regression coefficients, and ε is the residual error. Type II analyses of variance (ANOVAs) were computed to investigate the statistical significance of main effects of the three stressors on the response variables and their interaction effects (*car* package; Fox and Weisberg, 2019). In addition, standardized effect sizes (partial η^2 values, categories: < 0.10 very small, ≥ 0.10 small, ≥ 0.30 medium, ≥ 0.50 large) were calculated following (Nakagawa and Cuthill, 2007), to estimate the proportion of variance accounted for by each stressor combination (*lsr* package; v 0.5.1, Navarro and Navarro, 2022). For each response variable, the residuals were checked for normal distribution using Shapiro–Wilk's test, and for homoscedasticity using Levene's test (*rstatix* package; Kassambara, 2023). In the case of decomposition rate expressed per day during the recovery phase, the assumptions were only fulfilled when one outlier was removed. However, results obtained with and without the outlier included in the analyses were very similar and had no repercussion for the conclusions drawn.

Bray-Curtis dissimilarities were calculated on community data (raw spore counts) normalized by litter dry mass using the *vegan* package (Oksanen et al., 2022). The homogeneity of dispersion between treatments was tested using the *betadisper* function of the same package. A non-metric multidimensional scaling (NMDS) analysis was performed to assess the dissimilarities among aquatic hyphomycete communities among stressor combinations. To quantify the impact of stressors on community composition, a permutational multivariate analysis of variance (PERMANOVA) based on 999 permutations was performed using the same model structure as for the response variables (Eq. (1)) and the *adonis2* function of the *vegan* package (Oksanen et al., 2022).

To assess taxon-specific responses to the stressors, generalized linear models (GLM) were fitted to the raw spore counts (non-normalized) of all individual species accounting for at least 1 % of the total count across all mesocosms during both the stressor and recovery phase. The model included the three stressors as predictors as well as all two-way and three-way interactions. An offset term was added to account for variation in leaf mass. A quasi-Poisson regression with a logarithmic link-function was used because the model residuals showed overdispersion. Since the logarithmic link function of the GLM assesses stressor interactions on a multiplicative rather than an additive scale (Spake et al., 2023), a simple linear regression model according to Eq. (1) was applied as well. The same most abundant species were considered, with the raw spore counts divided by leaf mass used as the response variable this time.

For all models, the *emmeans* package (v 1.8.9., Lenth, 2023) was used to calculate estimated marginal means and their 95 % confidence intervals for all predictor terms that were significant ($p \leq 0.05$) or nearly significant ($0.05 < p \leq 0.10$).

3. Results

3.1. Aquatic hyphomycete community structure

Species richness of sporulating aquatic hyphomycetes ranged from 6 to 13 with an average of 11.1 ± 1.4 (SD) during the stressor phase and 9.5 ± 1.5 during the recovery phase (Fig. 2C, D, Table S4). No significant stressor effects on species richness were detected during the stressor phase (Table 1), but a nearly significant antagonistic interaction was apparent between temperature and salinity ($F_{1,31} = 3.46$, $p = 0.08$, partial $\eta^2 = 0.13$). Richness of sporulating aquatic hyphomycetes decreased at increased temperature ($t = 2.08$, $p = 0.05$, Fig. 3C) and salinity ($t = 1.84$, $p = 0.08$, data not shown), but the effect of both stressors in combination was lower than the sum of the single stressor effects. No interactive or individual effects of stressors on species richness were detected during the recovery phase (Table 1).

The community structure of sporulating aquatic hyphomycetes exposed to different stressor combinations did not clearly differ after either the stressor or the recovery phase (Fig. 5 B—I). This outcome of our NMDS analysis was supported by a PERMANOVA (Table S5) indicating a lack of effects except for a significant influence of reduced flow velocity during the recovery phase ($p = 0.03$; Table S5). Although not neatly grouped in two clusters in the NMDS analysis, communities slightly shifted between the end of the stressor and recovery phase ($p < 0.01$, Table S5; Fig. 5A).

3.2. Fungal sporulation

Averaged across all mesocosms, the sporulation rates of aquatic hyphomycetes were 1083 ± 578 (SD) spores mg^{-1} litter dry mass d^{-1} at the end of the stressor phase and 1080 ± 564 spores mg^{-1} d^{-1} after the recovery phase (Fig. 2A, B), with an overall range of 45 to 2563 spores mg^{-1} d^{-1}. None of the interactive or main effects was significant at the conventional 0.05 level (Table 1). However, although effect sizes were

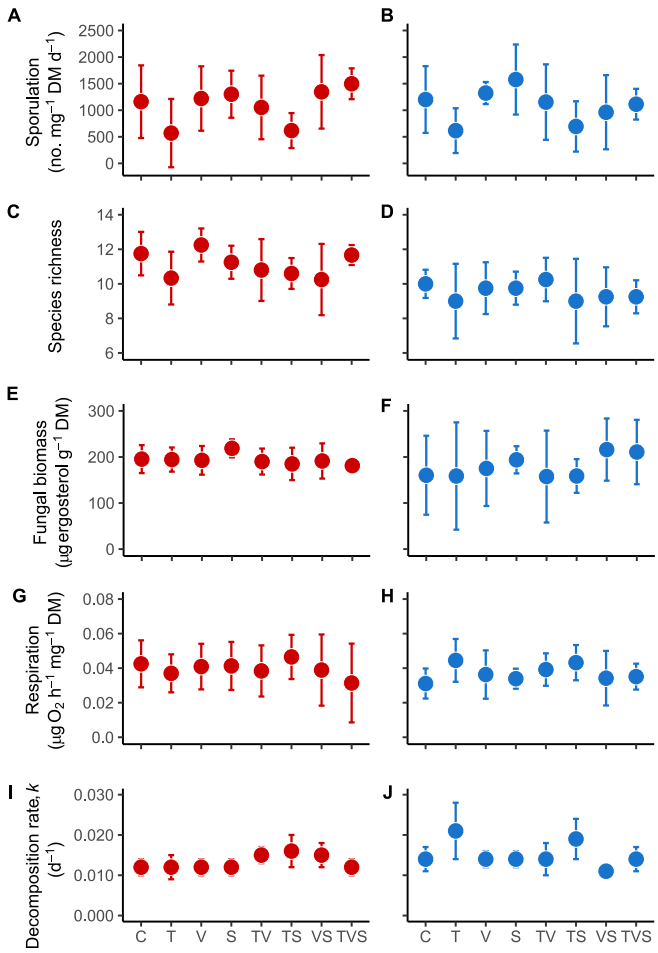

Fig. 2. Mean and standard deviation of the five responses variables, over the stressor phase (red) and the recovery phase (blue), each point corresponding to on treatment condition: Control (C), increased temperature (T), reduced flow velocity (V), increased salinity (S), increased temperature x reduced flow velocity (TV), increased temperature x increased salinity (TS), reduced flow velocity x increased salinity (VS), increased temperature x reduced flow velocity x increased salinity (TVS). A-B Sporulation rate, C—D Species richness, *E*-F Fungal biomass, G-H Respiration rate, I-J Litter decomposition rate. DM = leaf litter dry mass.

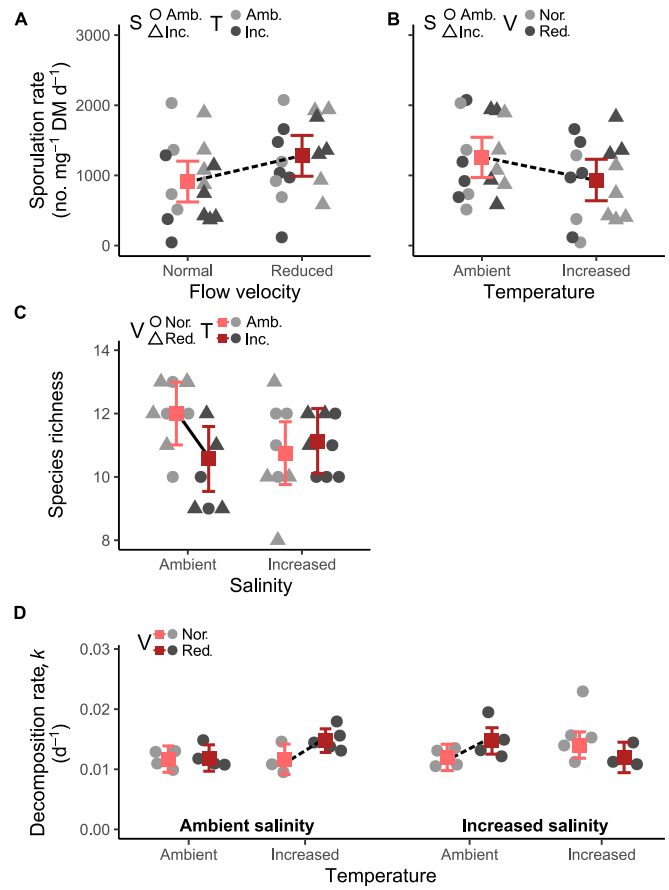

Fig. 3. Overview of significant ($p \leq 0.05$) and nearly significant ($0.05 < p \leq 0.1$) effects of studied response variables for the stressor phase. Marginal means from linear models and their 95 % confidence interval are plotted. A connecting line between the means indicates significance if solid ($p \leq 0.05$) and nearly significance if dashed ($0.05 < p \leq 0.1$). For two-way interactions, the design of the lines is based on post-hoc tests, with no line indicating a non-significant relationship. Raw data are represented by grey points, with the color and the shape of the points indicating their corresponding stressor conditions (T = Temperature, S = Salinity, V = Flow velocity, Amb. = Ambient, Inc. = Increased, Nor. = Normal, Red. = Reduced). DM = Leaf litter dry mass.

small (partial $\eta^2 = 0.11$ and 0.12, respectively), elevated temperature ($F_{1,31} = 2.89$, $p = 0.10$) resulted in a nearly significant decrease and reduced flow velocity ($F_{1,31} = 3.20$, $p = 0.09$) in a nearly significant increase in sporulation rate after the stressor phase (Fig. 3A, B). The negative influence of elevated temperature persisted until after the recovery phase (Fig. 4A), when a nearly significant antagonistic effect between temperature and flow velocity was detected as well ($F_{1,31} = 3.58$, $p = 0.07$, partial $\eta^2 = 0.13$; Fig. 4B). This interaction was due to decreased sporulation rates at elevated temperature under normal but not under reduced flow velocities.

Overall, the spore production by aquatic hyphomycetes across all mesocosms was dominated by four species, which accounted for 96.8 % of the total number of spores released during 24-h incubations after both the stressor and recovery phase. An unidentified species with sigmoid spores 60–70 µm in length sporulated most prolifically, followed by *Clavariopsis aquatica*, *Tetracladium marchalianum*, and *Amniculicola longissima*, in order of relative abundance (Table S6). In the stressor phase, increased temperature inhibited sporulation of *T. marchalianum* ($\chi^2_1 = 7.701$, $p < 0.01$, Fig. S1D) and *A. longissima* ($\chi^2_1 = 4.9$, $p = 0.03$; Fig. S1F). In contrast, reduced flow velocity stimulated sporulation of the dominant sigmoid species ($\chi^2_1 = 5.3$, $p = 0.02$; Fig. S1A) and for *A. longissima* ($\chi^2_1 = 4.9$, $p = 0.03$; Fig. S1E). Salinity only had a nearly significant effect and tended to stimulate the sporulation of *A. longissima* ($\chi^2_1 = 3.1$, $p = 0.08$; Fig. S1G). Furthermore, a synergistic interaction occurred between temperature and flow velocity for the sigmoid species ($\chi^2_1 = 2.7$, $p = 0.10$; Fig. S1B) and *A. longissima* ($\chi^2_1 = 7.1$, $p < 0.01$; Fig. S1H) with the stimulating effect of reduced flow velocity being enhanced by warming. However, the interactive effects were either not or poorly supported according to the results of the linear models (Table S6). Conversely, a nearly significant synergistic interaction occurred between flow velocity and salinity for the analysis of *A. longissima* sporulation using a linear model (Table S6). In the recovery phase, no effect of the stressors persisted. However, for the sigmoid species, sporulation rate tended to show context-dependent differences for mesocosms previously exposed to reduced flow velocity (three-way interaction, $\chi^2_1 = 2.7$, $p = 0.10$; Fig. S1C). In contrast to the effects of reduced flow velocity in the stressor phase, the same stressor reduced sporulation rates after the recovery phase, but only for the mesocosms that had been exposed to salinization and ambient temperature during the stressor phase (Fig. S1C).

3.3. Fungal biomass

Concentrations of ergosterol as an indicator of fungal biomass

Table 1
Results (F value, *p* values) of three-way ANOVAs for the response variables (sporulation rate, species richness, fungal biomass, respiration rates, leaf-litter decomposition rate) and for both phases (stressor and recovery). The predictors were temperature (T), flow velocity (V), salinity (S), temperature x flow velocity (TV), temperature x salinity (TS), flow velocity x salinity (VS), temperature x flow velocity x salinity (TVS). Values are printed in bold when $p \leq 0.05$, and in italic when $0.05 < p \leq 0.1$. The column *d* indicates with arrows the direction of the corresponding single stressor effect or the interaction type (A for antagonistic, S for synergistic) for the two-way interactions when significant or nearly significant.

Phase	Predictor	Sporulation rate (spores mg^{-1} litter dry mass d^{-1})			Species richness			Fungal biomass (μg ergosterol g^{-1} litter dry mass)			Respiration (mg O_2 h^{-1} mg^{-1} litter dry mass)			Litter decomposition rate *k* (d^{-1})		
		F	p	d	F	p	d	F	p	d	F	p	d	F	p	d
Stressor	T	*2.892*	*0.10*	↓	1.500	0.23		1.463	0.24		0.160	0.69		0.959	0.34	
	V	*3.204*	*0.09*	↑	0.246	0.63		0.849	0.37		0.495	0.49		1.933	0.18	
	S	0.892	0.35		0.577	0.46		0.014	0.91		0.008	0.93		0.755	0.39	
	TV	2.494	0.13		1.083	0.31		0.272	0.61		0.211	0.65		0.297	0.59	
	TS	0.081	0.78		*3.458*	*0.08*	A	0.905	0.35		0.079	0.78		1.460	0.24	
	VS	0.216	0.65		0.246	0.63		0.349	0.56		0.491	0.49		0.500	0.49	
	TVS	0.269	0.61		1.155	0.20		0.357	0.56		0.471	0.50		6.261	0.02	
Recovery	T	*3.768*	*0.06*	↓	0.314	0.58		0.290	0.60		2.567	0.12		6.920	0.02	↑
	V	0.366	0.55		0.113	0.74		0.628	0.44		0.735	0.40		7.226	0.01	↓
	S	0.005	0.94		0.615	0.44		1.328	0.26		0.002	0.97		0.653	0.43	
	TV	*3.579*	*0.07*	A	1.020	0.32		0.016	0.90		0.777	0.39		2.456	0.13	
	TS	0.001	0.98		0.013	0.91		0.036	0.85		0.460	0.50		0.085	0.77	
	VS	1.249	0.28		0.314	0.58		0.299	0.59		0.218	0.65		0.082	0.78	
	TVS	0.652	0.43		0.113	0.74		0.171	0.68		0.017	0.90		0.755	0.40	

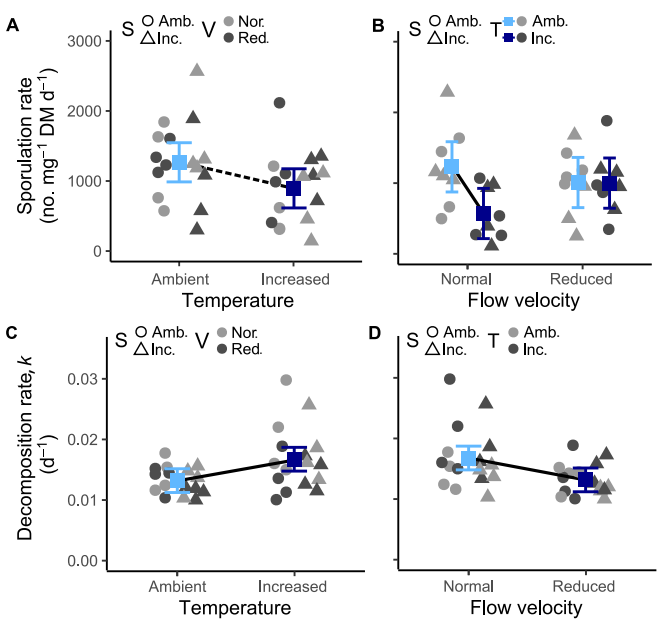

Fig. 4. Overview of significant ($p \leq 0.05$) and nearly significant ($0.05 < p \leq 0.1$) effects of studied response variables for the recovery phase. Marginal means from linear models and their 95 % confidence interval are plotted. A connecting line between the means indicates significance if solid ($p \leq 0.05$) and nearly significant if dashed ($0.05 < p \leq 0.1$). For two-way interactions, the design of the lines is based on post-hoc tests, with no line indicating a non-significant relationship. Raw data are represented by grey points, the color and the shape of the points indicating the stressor conditions. DM = Leaf litter dry mass.

averaged 194 ± 28 (SD) μg g^{-1} litter dry mass after the stressor phase, and 179 ± 73 μg g^{-1} after the recovery phase, with an overall range of 33 to 320 μg g^{-1} (Fig. 2E, F). During the stressor phase, concentrations varied within a relatively narrow range among mesocosms, but variation was much higher after the recovery phase, especially among replicate mesocosms (Fig. 2E, F). No stressor effects on fungal biomass were detected at the end of either the stressor or recovery phase (Table 1).

3.4. Microbial respiration

Microbial respiration rates ranged from 0.10 to 0.35 mg O_2 h^{-1} mg^{-1} litter dry mass. The average was 0.24 ± 0.01 (SD) mg O_2 h^{-1} mg^{-1} litter dry mass after the stressor phase, and of 0.22 ± 0.06 mg O_2 h^{-1} mg^{-1} after the recovery phase (Fig. 2G, H). No stressor effects on microbial respiration were detected at the end of the stressor or recovery phase (Table 1).

3.5. Leaf litter decomposition

Rates of leaf litter decomposition, expressed as exponential decay rate coefficient per day (*k*), varied more than threefold between 0.0095 and 0.0301 d^{-1}. The average rates were 0.0133 ± 0.0029 d^{-1} (SD) at the end of the stressor phase (i.e. after 38 days) and 0.0152 ± 0.0045 d^{-1} at the end of the recovery phase (i.e. after 53 days) (Fig. 2I, J). This corresponded to an average mass loss of 39.7 ± 6.1 % and 54.5 ± 9.1 % of the initial litter dry mass after the stressor and recovery phase, respectively. A significant but weak three-way interaction effect of all stressors was noted at the end of the stressor phase ($F_{1,30} = 3.26$, $p = 0.02$, partial $\eta^2 = 0.21$; Fig. 3D). Reduced flow velocities tended to stimulate decomposition, but only if it co-occurred with one or the other additional stressor. (Fig. 3D). By the end of the recovery phase, this three-way interaction effect was no longer apparent (Fig. 3C, Table 1). Instead, leaf litter decomposition was slowed in mesocosms formerly exposed to reduced flow velocity during the stressor phase ($F_{1,31} = 7.22$, $p = 0.01$), whereas formerly elevated temperature accelerated decomposition ($F_{1,31} = 6.92$, $p = 0.02$) (Fig. 4C, D), although both effects were small as well (partial $\eta^2 = 0.22$ and 0.23).

4. Discussion

4.1. Limited response of microbial decomposers to multiple stressors

The most conspicuous outcome of our controlled multi-stressor experiment in outdoor mesocosms mimicking restored urban streams is the limited response of microbial decomposers associated with leaf litter when exposed to single stressors or stressor combinations. The remarkable lack of responses detected for all three stressors applied in our experiment (i.e. warming, salinization and flow velocity reduction) was reflected by an array of variables, ranging from fungal community composition to quantitative measures of fungal presence and microbial activity, and ultimately to rates of microbial leaf litter decomposition, a

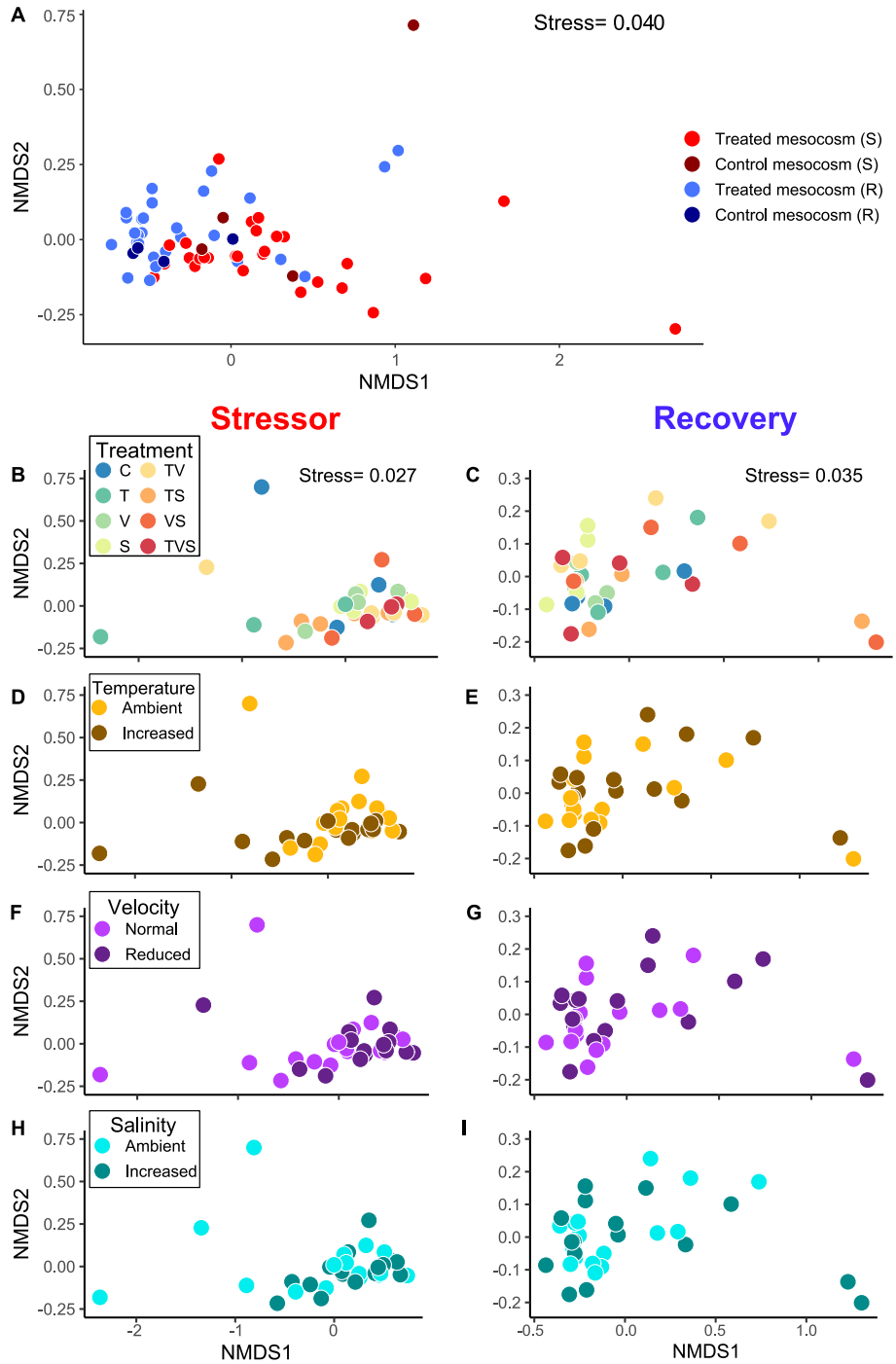

Fig. 5. Non-metric multidimensional scaling (NMDS) plots based on Bray-Curtis dissimilarities based on raw spore counts corrected by leaf mass. A: Based on the entire dataset (both stressor and recovery phases) and colored according to the sampling phase and to the treatment applied or not in the mesocosms (control or treated mesocosm). B-I: NMDS conducted separately on the different phases, with the color of the points depending on the mesocosm treatment: (B-C), the temperature treatment (D-E), the velocity treatment (F-G) or the salinity treatment.

critical ecosystem process in streams. The overall weak response we observed may be partly due to the short duration of stressor exposure compared to other mesocosm studies (e.g. Bruder et al., 2016; Nuy et al., 2018), pointing to the importance of repeated surveys (Juvigny-Khenafou et al., 2020; Juvigny-Khenafou et al., 2021). However, given that our flow-through mesocosms were primarily colonized by organisms from an urban stream that had been heavily impacted by coal mining and wastewater discharge and was only recently restored, it is also possible that the weak responses reflect a legacy of past exposure to high stressor levels leading to a "truncated" community, as has been reported from other impaired ecosystems (Feckler et al., 2018; Jackson et al., 2021). Although aquatic hyphomycetes were originally thought to be primarily associated with natural forest streams experiencing little impact of human activities (Bärlocher, 2016), these fungi have been frequently found to cope with environmental stress (Pascoal and Cássio, 2004; Krauss et al., 2011; Salis et al., 2023). Therefore, the truncated-community hypothesis could indeed apply to the litter-colonizing fungi in our mesocosms, the Boye River itself, and possibly heavily impacted urban streams in general.

The strong dominance of the communities by a single fungal species

in our study and only three subdominant species is a possible factor limiting opportunities for shifts in community composition and relative species abundances in response to changing environmental conditions. However, fungal biomass, estimated at about 3 % of litter dry mass if a conversion factor of 5 μg ergosterol per mg fungal dry mass is applied (Brosed et al., 2017), was not exceedingly low, although lower than often found in streams in relatively labile types of leaf litter such as alder leaves (Gessner, 2020; Schindler and Gessner, 2009). Other response variables were also within the range of published data, including (i) the sporulation rate of aquatic hyphomycetes (Bärlocher, 2020), which tends to be the most responsive measure of microbial activity in decomposing leaf litter exposed to anthropogenic stressors in streams (Lecerf and Chauvet, 2008), (ii) rates of litter-associated microbial respiration (Graça and Abelho, 2020), which proved to be as resistant to stressor exposure as fungal biomass, and (iii) litter decomposition rates (e.g. Brosed et al., 2022). These comparisons show that fungi were clearly active and promoted considerable litter decomposition (average of 40 % litter dry mass loss at the end of the stressor phase), suggesting that a lack of microbial decomposer activity does not readily explain the paucity of stressor effects in our experiment.

4.2. Temperature relationships

Both previous field observations and experiments in microcosms and streams indicate that temperature plays an important role in structuring aquatic hyphomycete communities developing in decomposing leaf litter (Gonçalves et al., 2013). In particular, community shifts involving species replacements are expected when water temperature exceeds 16–18 °C (Gessner et al., 1993; Suberkropp, 1984), which is much higher than the 3.5 °C increase above the mean ambient temperature of 8.7 °C during our experiment. Nevertheless, the nearly significantly reduced total sporulation rates on leaf litter from our heated mesocosms corroborate field data suggesting that communities may still respond to warming below the 16–18 °C threshold (Gessner et al., 1993).

Although weak as well effects of warming by 3.5 °C were indeed the strongest among all stressor effects in our experiment, either alone or in combination with salinization or reduced flow velocity. Temperature stimulates both sporulation (Chauvet and Suberkropp, 1998; Ferreira and Chauvet, 2011b) and respiration of aquatic hyphomycetes in laboratory conditions (Jabiol et al., 2020). However, such temperature effects are not always clear in the field (Ferreira and Canhoto, 2015) and can even be negative (Martínez et al., 2014), as also seen in our experiment, where sporulation rates were determined at a standard temperature. A possible reason for this counterintuitive result is that the sporulation potential of aquatic hyphomycetes changes markedly during fungal succession in decomposing leaf litter (Bärlocher, 1982; Gessner et al., 1993). If temperature accelerates succession such that the sporulation peak is surpassed, sporulation in heated mesocosms may be less than at ambient temperature. Chances that such a situation arises are particularly high when samples are taken at a single time point when litter mass loss exceeds 50 % (Gessner and Chauvet, 1994). That stage had not yet been reached by the end of the stressor phase in our experiment when an average of about 40 % litter dry mas was remaining. Nonetheless, decomposition may have sufficiently proceeded by that time to make it hard to detect effects, especially in view of the considerable variability associated with the measurement of sporulation rates.

4.3. Remarkable tolerance to elevated salinity

The influence of temperature on fungal decomposers could be modified by other stressors, as indicated by the nearly significant temperature effect on species richness at ambient but not elevated salinity. Other salinization effects were virtually undetectable for any of the measured response variables. This finding is consistent with published results of microcosm and field studies (Canhoto et al., 2017; Almeida Júnior et al., 2020; Gonçalves et al., 2019; García et al., 2024), suggesting a pronounced tolerance of aquatic hyphomycetes to salinization well beyond the freshwater limit of 1 ‰. Some species, including one of the four common ones in our mesocosms, *T. marchalianum*, can even maintain growth in media at NaCl concentrations of up to 16 g L^{-1} (Canhoto et al., 2017). Thus, although sporulation tends to respond more sensitively to salinization than growth (Canhoto et al., 2021), the unresponsiveness in our experiment of both metrics is not surprising. Indeed, there is a discrepancy between the salinity level we tested (< 1 g L^{-1} NaCl added) and the concentrations generally reported to have adverse effects. For instance, effects on the sporulation of aquatic hyphomycete cultures in microcosms required the addition of 2 g L^{-1} NaCl (Canhoto et al., 2017) or even up to 8 g L^{-1} (Pereira da Silva et al., 2021). Similarly, a test of effects on the growth rates of aquatic hyphomycetes on agar plates resulted in an $EC_{50} \geq 8$ g L^{-1} (Canhoto et al., 2017).

As expected in view of the tolerance of aquatic hyphomycetes, microbial leaf litter decomposition has also been found to be little affected by elevated salinity in streams, including in the present study, even at relatively high levels (Canhoto et al., 2021). Conversely, reduced rates of leaf litter decomposition and species richness of aquatic hyphomycetes have been found at elevated salinity in streams receiving mining effluents (Lecerf and Chauvet, 2008), although rates remained unaffected in other studies at conductivities of up to nearly 1400 μS cm^{-1} (Vander Vorste et al., 2019). Effects in these mining streams, however, were more likely caused by exposure to heavy metals and/or low pH than by increased salinity (Lecerf and Chauvet, 2008; Niyogi et al., 2002).

To summarize, a major reason for the lack of salinization effects on microbial decomposers and leaf litter decomposition in our experiment appears to be a remarkable resistance of aquatic hyphomycetes to salt stress (Canhoto et al., 2021). The high background salinity (>800 μS cm^{-1}; Diamantini et al., 2018; Madge Pimentel et al., 2024) in the Boye River supplying water and fungal propagules to our mesocosms implies a legacy of past stressor exposure in this recently restored urban stream, which manifested as pollution-induced community tolerance (Tlili et al., 2016). The importance of this mechanism is underlined by a case study showing higher sporulation rates of aquatic hyphomycete strains isolated from a salt-rich stream when exposed to elevated salinities compared to strains from a low-salinity reference stream (Gonçalves et al., 2019).

4.4. Little evidence of flow velocity effects

Adverse effects of flow reduction on litter-associated fungal biomass, respiration, sporulation and sometimes leaf litter decomposition have been reported from both field and laboratory settings (Ferreira and Graça, 2006; Schlief and Mutz, 2011; Bruder et al., 2016; Juvigny-Khenafou et al., 2020). Potential mechanisms underlying such effects include low dissolved oxygen concentrations limiting the metabolic activity of aquatic hyphomycetes (Medeiros et al., 2009), smothering of leaf surfaces by settled suspended matter, potentially hampering litter colonization and/or spore release (Schlief and Mutz, 2009), and the stimulation of sporulation by turbulence (Ferreira and Graça, 2006). However, as for salinity, flow velocities in our experiment (0.1 vs 0.2 m s^{-1}) were invariably within the range of typical baseflow conditions in the field, explaining the overall unresponsiveness observed in our experiment.

4.5. Recovery, delays and other temporal effects

An important aim of our experiment was to determine the recovery potential of microbial decomposers and their influence on leaf litter decomposition after being released from the impact of stressors, as set out in the Asymmetric Response Hypothesis of Vos et al. (2023). However, a prerequisite to testing for recovery is the demonstration of prior

stressor impact, a requirement that was not met in our experiment. Effects emerging after the recovery phase are therefore best interpreted as delayed impacts of previous stressor exposure. For instance, the nearly significant effects of temperature and flow velocity on sporulation became an interactive effect, showing that sporulation can continue to be impacted even after the release from these stressors. Moreover, a positive effect on leaf litter decomposition by warming and a negative effect of reduced flow velocity occurred after stressor removal. These results could point to some potential of temperature, the strongest among three weak stressors in our experiment, to affect aquatic hyphomycete communities and microbial litter decomposition in streams, corroborating other experimental results that show temperature increases within a certain range to stimulate leaf litter decomposition (Fernandes et al., 2012; Ferreira and Chauvet, 2011a; Martínez et al., 2014). Moreover, the slower decomposition observed in our mesocosms previously exposed to reduced flow velocity also hint at a delayed influence of altered flow velocity.

Despite the limited impacts of the three stressors tested in our study, community structure changed between the end of the stressor and recovery phase. Community structure can predictably shift between early and advanced stages of leaf litter decomposition (Bärlocher, 1992; Gessner et al., 1993), presumably because fast growing and prolifically sporulating species exploiting labile leaf resources are gradually supplanted by slow-growing species exploiting increasingly refractory leaf tissue and reproducing less vigorously. It is important to note that these dynamics mainly reflect different stages of fungal succession and litter decomposition (Bärlocher, 1992; Gessner et al., 1993), rather than delayed stressor effects or recovery.

5. Conclusion

Overall, our mesocosm experiment in a recently restored urban stream revealed that leaf litter decomposition and litter-associated fungi did not strongly respond to any of the three stressors tested: warming, salinization, and reduced flow velocity. However, despite the overall scarcity of effects, the delayed decrease in leaf litter decomposition rates after previous exposure to warming and reduced flow velocity highlight potential challenges in the management of urban streams recovering from multiple stressors. Impacts observed after restoration, apart from being due to new stressors, could reflect impairments that might persist for decades after stressor removal (Jähnig et al., 2011; Gillmann et al., 2023). If such legacies include changes in rates of litter decomposition, they could induce shifts in the temporal distribution of leaf litter over the year. This, in turn, could alter the availability of food resource in relation to macroinvertebrate life histories, with indirect consequences for other components of the biological communities and ecosystem processes in urban streams.

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CRediT authorship contribution statement

Gwendoline M. David: Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft. **Iris Madge Pimentel:** Formal analysis, Investigation, Methodology, Project administration, Writing – review & editing. **Philipp M. Rehsen:** Investigation, Methodology, Project administration, Writing – review & editing. **Anna-Maria Vermiert:** Investigation, Methodology, Project administration, Writing – review & editing. **Florian Leese:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing. **Mark O. Gessner:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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